

Warm-up

(1) Does $\sum_{n=1}^{\infty} \frac{n}{n^4 + 1}$ converge or diverge?

(2) Does $\sum_{n=1}^{\infty} \frac{n^2 - 3}{n^3 + 2}$ converge or diverge?

Medium

Determine whether the following series converge or diverge:

$$(1) \sum_{n=1}^{\infty} \frac{n^3 + n - 1}{n^5 - 3n^4} \quad (2) \sum_{n=1}^{\infty} n e^{-n^2+1} \quad (3) \sum_{n=2}^{\infty} \frac{\ln(n) + n}{n^2 - 1} \quad (4) \sum_{n=1}^{\infty} \frac{n! + 1}{(n+1)!}.$$

Stretch

(1) We can use an argument like the DCT to show that the harmonic series diverges. Note that the first two terms 1 and $\frac{1}{2}$ are both at least $\frac{1}{2}$, and similarly the next two terms add to something greater than $\frac{1}{2}$. Continue collecting the terms of the harmonic series into groups that are at least $\frac{1}{2}$ to show that the harmonic series is greater than $\frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \cdots = \infty$.

(2) Use a grouping method similar to the above to show that for any non-increasing sequence (a_n) , we have that

$$\sum_{n=1}^{\infty} a_n \leq \sum_{n=1}^{\infty} 2^n a_{2^n} \leq \sum_{n=1}^{\infty} 2a_n.$$

Conclude by the direct comparison test that $\sum_{n=1}^{\infty} a_n$ converges if and only if $\sum_{n=1}^{\infty} 2^n a_{2^n}$ does.

(This result is known as the *Cauchy condensation test*.)